

Deep Reinforcement Learning-Enabled Anti-Reactive Jamming Strategies for Mobile Cell-Free Interference Networks

Tri Nhu Do[†], Georges Kaddoum^{*}, and François Leduc-Primeau[‡]

^{†,‡}Department of Electrical Engineering, Polytechnique Montréal, Montreal, QC, Canada

^{*}Department of Electrical Engineering, École de technologie supérieure (ÉTS), University of Quebec, Montreal, QC, Canada
Emails: [†]tri-nhu.do@polymtl.ca, ^{*}georges.kaddoum@etsmtl.ca, [‡]francois.leduc-primeau@polymtl.ca

Abstract—In this paper, we propose a new concept of a mobile cell-free interference network and address the collision-free frequency allocation (CFFA) problem in this network under reactive jamming. The CFFA is formulated as a mixed-integer optimization problem with finite state and action spaces, which can be solved using exhaustive search, assuming the availability of the jammer’s channel state information (CSI). Considering a more realistic scenario in which the jammer’s CSI is unknown prior to the attack, we reformulate the CFFA as an interactive Markov decision process and propose a double deep Q-learning (DDQL) approach to minimize the loss caused by the jammer and achieve CFFA for the remaining active transmissions. Within this framework, the network operates as a cognitive radio system that attains real-time situational awareness through DDQL interactions, enabling adaptive responses to dynamic jamming behaviors. By considering reference point group mobility for legitimate users and random walk mobility for the jammer, we demonstrate that the proposed DDQL achieves a near-optimal policy compared to the optimal exhaustive search. We conclude that, in the presence of unavoidable active jamming, there exists a policy for multiple concurrent cell-free transmission pairs that can minimize spectral efficiency loss in a mobile environment.

Index Terms—Interference channel, cell-free network, spectral efficiency optimization, reinforcement learning, deep Q-learning

I. INTRODUCTION

A mobile cell-free interference (MCFI) network is a dynamic communication architecture where a large number of mobile transmitter–receiver pairs operate concurrently, forming an interference network. This network inspired by the conventional interference channel [1] and the principle of cell-free MIMO [2], but generalized to arbitrary interference networks, rather than joint transmission from co-located/distributed antennas. A potential application of the MCFI framework is in tactical wireless networks. In the contested environment of these networks, active jamming attacks are among the most critical problems [3]. Moreover, developing efficient countermeasures remains a challenging problem in the literature [4].

Recently, several machine learning-based counterattack strategies have been proposed. In [5], the authors proposed a

method to combat jamming attacks on Physical Uplink Control Channels (PUCCHs) in the Third Generation Partnership Project (3GPP) Long Term Evolution (LTE) for either public safety or military use cases. In [4], the authors proposed a deep reinforcement learning (DRL)-enabled anti-reactive jamming attack for a simple network consisting of one transmission pair under the impact of a reactive jammer. In [6], the authors relied on game theory to address intelligent radio frequency (RF) jamming attacks on cognitive radio networks. In [7], the authors used reinforcement learning (RL) to improve conventional methods, such as the energy detector, for jamming sensing in electromagnetic spectrum warfare scenarios.

In this paper, we consider a more realistic scenario of such a network, which comprises two independent parties: legitimate users and a jammer, which operate independently and against each other. Since neither side has knowledge of the other, the key point is that there are two observations associated with one physical entity, e.g., two channel coefficients associated with one physical channel, since it is viewed differently by legitimate users and the jammer. In addition, we consider a practical situation where reactive jamming cannot be avoided because the attack is only detected after the network has suffered performance loss. To address this problem, we employ a multi-frequency hopping tactic integrated within a *cognitive radio framework*, where the network exhibits real-time situation awareness (SA), the capability to continuously perceive the surrounding spectrum environment, comprehend the ongoing jamming state, and issue an adaptive hopping policy.

We first design a collision-free frequency allocation (CFFA) strategy for the considered network and then formulate it as a stochastic programming optimization problem, where the behavior of the jammer, i.e., the next channel being jammed, is stochastic and unknown. Through this cognitive layer, the system achieves adaptive decision-making via RL interaction, where the RL agent serves as the cognitive agent. The formulated problem is then characterized as a sequential decision-making problem and converted to a Markov decision process (MDP). To solve the formulated MDP, we resort to deep Q-network (DQN)-based reinforcement learning, through which the cognitive agent continuously refines its situation

awareness and adapts to jamming dynamics. We first propose a double DQN (DDQN) algorithm to solve the MDP. Major contributions of the present study include the following¹:

- We design a novel anti-reactive jamming attack strategy, called CFFI, for a jammed mobile cell-free interference network, which can minimize performance loss and achieve maximum unjamming-effective SE.
- Considering that the channel state information (CSI) of the jammer is unknown, we convert the formulated optimization problem to an MDP and propose a DDQN algorithm to solve the formulated MDP. The obtained optimal policy is validated by exhaustive search, which confirms its correctness.
- Our numerical results demonstrate that the proposed DDQN algorithm successfully find a near-optimal solution to the original optimization problem. The obtained solutions can be validated since they provide the same performance pattern as that obtained by the optimal solution found by the exhaustive search algorithm (note that this search requires full jamming knowledge).

II. NETWORK DESCRIPTION AND SIGNAL MODELING

We consider a MCFI network consisting of a set of N single-antenna user pairs, i.e., (T_n, R_n) , $n = 1, \dots, N$, sharing a set of K orthogonal frequency bands, $\mathcal{F} = \{f_k, k = 1, \dots, K\}$, and $K \geq (N + 1)$. We consider that all nodes in the MCFI are equipped with a single antenna. The legitimate network is coordinated by a central entity, called the Hub and denoted as H. We consider the MCFI as a backlogged network [8], i.e., all N transmitters have data to send in each time block. There is a presence of a reactive jammer, J, that monitors and launches attacks on the legitimate pairs. It is noted that there are two wireless channels associated with one frequency band f_k , namely the legitimate channel h and the channel from this transmitter to the jammer.

A. Time Division Structure and Channel Modeling

We assume that the time domain is divided into equal-length time blocks. Considering quasi-static fading, a channel coherence time block t , with the interval of T_c , is divided into Λ equal sub-time steps t_τ , $\tau = 1, \dots, \Lambda$, each with an interval of T_c/Λ . In each time block, the fading is quasi-static within a channel coherence time block T_c . The complex channel coefficient of the transmission from node A to node B, where $A, B \in \{H, T_n, R_n, J\}$, over frequency band f_k is $\tilde{h}_{AB}^{f_k}$, and its real-valued magnitude is modeled as $|\tilde{h}_{AB}^{f_k}| = \sqrt{\mathcal{L}_{AB}} h_{AB}^{f_k}$, where $\mathcal{L}_{AB}$ [dB] represents the large-scale fading, i.e., the path-loss in our MCFI, i.e., $1/d^{\text{ple}}$, where ple denotes the path-loss exponent. f_c [GHz] denotes the carrier frequency, d_{AB} [m] is the distance between nodes A and B in a two-dimensional Cartesian coordinate system, and d_0 [m] is the reference distance; and $h_{AB}^{f_k}$ represents the small-scale fading of the channel. Considering Nakagami- m fading, i.e., the

magnitude of the channel coefficient follows a Nakagami- m distribution, parameterized by m and Ω , with the PDF given by $f_X(x; m, \Omega) = \frac{2m^m}{\Gamma(m)\Omega^m} x^{2m-1} \exp(-\frac{m}{\Omega}x^2)$, where $m > 0$ is the *shape parameter*, indicating the severity of fading, and $\Omega > 0$ is the *spread parameter* of the distribution, indicating the average channel (power) gain. Without loss of generality, we assume that channels are reciprocal, i.e., $\tilde{h}_{XY}^{f_k} = \tilde{h}_{YX}^{f_k}$. For a given set of frequency bands, $\mathcal{F} = \{f_1, \dots, f_K\}$, the channel matrix associated with legitimate pairs, $\mathbf{H} \in \mathbb{R}^{N \times K}$, and the channel matrix associated with the jammer, $\mathbf{G} \in \mathbb{R}^{N \times K}$.

B. The Proposed Anti-Reactive Jamming CFFA Strategy

1) *Traffic Channel Access and Data Transmission*: At time step t_τ , each transmission $T_n \rightarrow R_n$ is conducted over a frequency band $f_{D,n}[t_\tau] \in \mathcal{F}$, which is sent by the Hub in the broadcasted strategy. It is noted that $f_{D,n}[t_\tau]$ and $f_{D,n}[t_{\tau+1}]$ are statically independent. Assuming T_n is transmitting with power P_{T_n} , let x_{T_n} , where $\mathbb{E}[|x_{T_n}|^2] = 1$, denote the transmitted signal, the received signal at R_n can be expressed in case with CFFI as

$$y_{T_n R_n}^{f_{D,n}}[t_\tau] = \sqrt{P_{T_n}} \sqrt{\mathcal{L}_{T_n R_n}^{f_{D,n}}} \tilde{h}_{T_n R_n}^{f_{D,n}} x_{T_n}[t_\tau] + n_{R_n}[t_\tau], \quad (1)$$

where n_{R_n} denotes the additive white Gaussian noise (AWGN) at R_n , i.e., $n_{R_n} \sim \mathcal{CN}(0, \sigma_{R_n}^2)$, with $\sigma_{R_n}^2$ [dBm] = $N_0 + 10 \log_{10}(\text{BW}) + \text{NF}$, where N_0 is the noise power spectral density [dBm/Hz], BW is the signal bandwidth [Hz], and NF is noise figure [dB]. The instantaneous received signal-to-noise ratio (SNR) at R_n over frequency band $f_{D,n}$ at time step t_τ , $\text{SNR}_{T_n R_n}^{f_{D,n}}$, can be expressed as $\text{SNR}_{T_n R_n}^{f_{D,n}}[t_\tau] = P_{T_n} \mathcal{L}_{T_n R_n}^{f_{D,n}} |\tilde{h}_{T_n R_n}^{f_{D,n}}|^2 / \sigma_{R_n}^2$. The instantaneous achievable SE at R_n over frequency band $f_{D,n}$ at time step t_τ , $R_{T_n R_n}^{f_{D,n}}$ [b/s/Hz], can be expressed as

$$R_{T_n R_n}^{f_{D,n}}[t_\tau] = \log_2(1 + \text{SNR}_{T_n R_n}^{f_{D,n}}[t_\tau]). \quad (2)$$

There are two reasons for the failure of a tactical transmission, being (i) the TCH $f_{D,n}$ is jammed or (ii) there is a collision on the TCH $f_{D,n}$, i.e., there is a transmitter T_j with $f_{D,n} \equiv f_{D,j}$. Thus, the data transmission is successful if T_n is able to receive an acknowledgment (ACK) for its transmission at the end of each time step t_τ , which can be expressed as

$$\mathbb{I}(f_{D,n}[t_\tau]) = \begin{cases} 1, & \text{if } f_{D,n}[t_\tau] \neq f_J[t_\tau], \\ & \text{and } f_{D,n}[t_\tau] \neq f_{D,j}[t_\tau], \forall i, \forall j, \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

if $f_{D,n}[t_\tau]$ is jamming-free and no collision.

2) *Jamming Detection and Reporting*: The reactive jammer launches attacks on active transmissions, i.e., after the tactical transmitters start their transmissions. Before that, the reactive jammer hides itself by keeping silent. To detect the jammer, the transmitter T_n relies on the concept of congestion probing, also known as ACK probing [9]. The operation of the jammer is determined by the Hub through the following two events: (i) a TCH being attacked and (ii) collided TCHs. As will be explained in the next section, the CFFA is designed to issue a

¹The code repository for this paper, with more demonstrations, is available at <https://github.com/TND-lab/DRL-Active-Jamming-Interference-Networks>

collision-free channel allocation strategy, and the Hub focuses solely on the jamming status of frequency bands in $\mathcal{F}$. If a data channel is attacked, the associated pair cannot report, while other pairs can still report, and the Hub decides that the channel without the report is jammed. Finally, based on the collected messages, the Hub issues a CFFA policy $\pi[t_{\tau+1}]$ for the next time step $t_{\tau+1}$.

C. Operation and Modeling of the Reactive Jammer

Recalling that, in this work, we consider a reactive jamming attack, where the reactive jammer launches an attack only if it detects active transmissions. Specifically, we assume that the jammer is capable of continuously and simultaneously monitoring all K frequency bands, and attacks only the strongest active transmission, i.e., the one with the highest observed SNR at the jammer. The overheard signal, $y_{T_n J}^{f_{J,n}}$, associated with the $T_n \rightarrow R_n$ transmission observed at the jammer can be expressed as

$$y_{T_n J}^{f_{J,n}}[t_\tau] = \sqrt{P_{T_n}} \sqrt{\mathcal{L}_{T_n J}^{f_{J,n}}} \tilde{h}_{T_n J}^{f_{J,n}} x_{T_n}[t_\tau] + n_J[t_\tau]. \quad (4)$$

where $f_{J,n}[t_\tau]$ denotes the overheard channel from T_n to J . The overhead SNR, $\text{SNR}_{T_n J}^{f_{J,n}}$, associated with the $T_n \rightarrow R_n$ transmission at the jammer can be expressed as $\text{SNR}_{T_n J}^{f_{J,n}}[t_\tau] = P_{T_n} \mathcal{L}_{T_n J}^{f_{J,n}} |\tilde{h}_{T_n J}^{f_{J,n}}|^2 / \sigma_J^2$. The operations of the jammer and the legitimate pairs are independent and asynchronous. Specifically, the indicator of the presence of the jammer on frequency band f_k , $\mathbb{I}_{J,k}$, is expressed as

$$\mathbb{I}_{J,k}[t_\tau] = \begin{cases} 1, & \text{if data transmission,} \\ 0, & \text{if no active transmission.} \end{cases} \quad (5)$$

The jammer chooses to attack a frequency band, denoted by f_J^* , that has the highest overheard SNR, which can be mathematically expressed as

$$f_J^*[t_\tau] = \arg \max_{n=1,\dots,N} \text{SNR}_{T_n J}^{f_{J,n}}[t_\tau]. \quad (6)$$

When a higher active energy level is detected on another transmission, the jammer immediately switches to the new target attack transmission.

III. SPECTRAL EFFICIENCY MAXIMIZATION AND MARKOV DECISION PROCESS PROBLEM FORMULATION

A. Spectral Efficiency Maximization Problem Formulation

Let π_{t_τ} denote the CFFA strategy issued by the Hub for time step t_τ . Let $\mathbf{f}^{\pi_{t_\tau}}$ denote the vector of the indices of frequency bands associated with the strategy π_{t_τ} , i.e., $\mathbf{f}^{\pi_{t_\tau}} = [f_{D,1}^{\pi_{t_\tau}}, \dots, f_{D,n}^{\pi_{t_\tau}}, \dots, f_{D,N}^{\pi_{t_\tau}}]$. Based on the proposed transmission scheme and the operation of the reactive jammer, the instantaneous spectral efficiency (SE) of the network, SE_Σ [b/s/Hz], can be expressed as

$$\text{SE}_\Sigma[t_\tau] = \sum_{n=1}^N R^{f_{D,n}^{\pi_{t_\tau}}}[t_\tau] \mathbb{I}(f_{D,n}^{\pi_{t_\tau}}[t_\tau]), \quad (7)$$

where $R^{f_{D,n}^{\pi_{t_\tau}}}$ is defined in (2), where $R^{f_{D,n}}[t_\tau] \mathbb{I}(f_{D,n}^{\pi_{t_\tau}}[t_\tau]) = 0$ if $f_{J,n}^{\pi_{t_\tau}} = f_J^*$. The objective is to find the optimal CFFA policy,

$\pi_{t_\tau}^*$, such that the maximum instantaneous SE is achieved, i.e.:

$$\underset{\mathbf{f}^{\pi_{t_\tau}}}{\text{maximize}} \quad \text{SE}_\Sigma, \quad (8a)$$

$$\text{subject to} \quad f_{D,n}^{\pi_{t_\tau}} \neq f_{D,l}^{\pi_{t_\tau}}[t_\tau], \forall n, \forall l, \forall t_\tau, \quad (8b)$$

$$|\mathcal{F}_J[t_\tau]| = 1, \forall t_\tau, \quad (8c)$$

$$R^{f_{D,n}^{\pi_{t_\tau}}}[t_\tau] = 0 \text{ if } \mathbb{I}(f_{D,n}^{\pi_{t_\tau}}[t_\tau]) = 1, \forall t. \quad (8d)$$

The key technical challenge behind the formulated optimization problem is that the problem in (8) is a discrete optimization problem since the optimization variable vector takes values from a discrete set, i.e., the indices of the frequency bands in $\mathcal{F}$. Thus, problem (8) can be categorized as a *pure integer programming problem*, which is a special case of the *mixed-integer nonlinear programming problem* (MINLP). In addition, due to the unique ability of the reactive jammer, i.e., attacking after the legitimate transmission starts, the information of a jammed frequency band is information about the future and cannot be known by the Hub with certainty.

B. Equivalent Markov Decision Process Problem Formulation

To address the problem in (8), we convert it to a Markov decision process problem, as presented next.

1) *State Space*: The state space $\mathcal{S}$ is the set of all possible states, which can be expressed as

$$\mathcal{S} = \{\mathbf{s}_l : l = 1, 2, \dots, |\mathcal{S}|\}, \quad (9)$$

where $1 \times K$ vector $\mathbf{s}_l$ represents the l -th state in the state space, which can be expressed as

$$\mathbf{s}_l = [s_{f_1}, \dots, s_{f_k}, \dots, s_{f_K}, f_J^*], \quad (10)$$

where $[s_l]_1 = s_{f_1}$, $[s_l]_k = s_{f_k}$, $[s_l]_K = s_{f_K}$, $[s_l]_{K+1} = f_J^*$. Note that the agent does not use the jammer's CSI to determine f_{jammer}^* , instead, it is inferred from observations made after the attack. Consequently, $\mathbf{s}[t_\tau] \in \mathcal{S}$ means the state of the MCFIs at time step t_τ , where s_{f_k} is a non-numeric element representing the in-use status of frequency band f_k at time step t_τ , defined as

$$s_{f_k} = \begin{cases} s_{f_k} \in \{u1, \dots, uN\}, & \text{if it is used by } T_n, \\ \emptyset, & \text{if it is unused.} \end{cases} \quad (11)$$

2) *Action Space*: The action space $\mathcal{A}$ is the set of all possible actions that can be performed by the agent, which can be expressed as

$$\mathcal{A} = \{\mathbf{a}_m : m = 1, 2, \dots, |\mathcal{A}|\}, \quad (12)$$

where the $(N+1) \times 1$ numeric column vector $\mathbf{a}_m$ is the m -th action in the discrete action space, which is defined as

$$\mathbf{a}_m = [f_{D,1}, \dots, f_{D,n}, \dots, f_{D,N}]^T, \quad (13)$$

where $f_{D,n} \in \{1, \dots, K\}$, $n \in \{1, \dots, N\}$ indicate the indices of the frequency band being assigned to T_n , respectively. Consequently, $\mathbf{a}[t_\tau] \in \mathcal{A}$ denotes the action the Hub issues at time step t_τ . To address the TCH collision problem, i.e.,

constraint (8b) in the formulated problem (8), the vectors $\mathbf{a}_m$ are generated as permutations without repetitions of elements from the set $\{f_{D,1}, \dots, f_{D,n}, \dots, f_{D,N}\}$.

3) *Reward*: Let $r : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \mapsto \mathbb{R}$ denote the reward function, which defines the *immediate reward* for executing action $\mathbf{a}$ and transitioning state $\mathbf{s}$ to state $\mathbf{s}'$. The reward is designed to maximize the instantaneous achievable SE, i.e.:

$$r(\mathbf{s}, \mathbf{a})[t_\tau] = \sum_{n=1}^N R_{f_{D,n}[t_\tau]|\mathbf{a}[t_\tau]} \quad (14)$$

4) *Transition Probability*: The transition from one state to another is characterized by the transition probability. Let $p : \mathcal{S} \times \mathcal{A} \mapsto [0, 1]$ denote the probability of transitioning from state $\mathbf{s}$ to state $\mathbf{s}'$ after executing an action $\mathbf{a}$, i.e., $p = \Pr(\mathbf{s}[t_{\tau+1}] = \mathbf{s}' | \mathbf{s}[t_\tau] = \mathbf{s}, \mathbf{a}[t_\tau])$. In our problem, by treating actions as hypotheses, the transition probability is a function of the legitimate posterior and the jamming likelihood, which can be expressed as

$$p(\mathbf{s}', r | \mathbf{s}, \mathbf{a}) = \Pr(\mathbf{a} | \mathbf{s}, \mathbf{H}) \Pr(f_J^* | \mathbf{G}) \Pr(f_{J,n} = f_J^*). \quad (15)$$

where $\Pr(\mathbf{a} | \mathbf{s}, \mathbf{H}) = \prod_{n=1}^N \Pr(f_{D,n} = f_k)$, $\mathbf{H}$ and $\mathbf{G}$ denote the legitimate and jamming CSI matrices, respectively. Note that $\mathbf{G}$ is unknown to the agent.

Remark 1. The considered CFFA problem can be formulated as a finite MDP. As a result, the SE optimization problem in (8) is converted to an MDP, which can be defined by a 4-tuple, i.e., $\langle \mathcal{S}, \mathcal{A}, p, r \rangle$, where $\mathcal{S}$, $\mathcal{A}$, p , and r are defined in (9), (12), (15), and (14), respectively.

A CFFA strategy is an output of a *policy*, i.e., $\pi : \mathcal{S} \rightarrow \mathcal{A}$, which is a function that maps a given state $\mathbf{s} \in \mathcal{S}$ to its corresponding action $\mathbf{a} \in \mathcal{A}$, i.e., $\pi(\mathbf{s}[t_\tau]) = \mathbf{a}[t_\tau]$. Solving problem (8) is equivalent to finding the optimal policy for the formulated MDP $\langle \mathcal{S}, \mathcal{A}, p, r \rangle$, so that, at an arbitrary time step t_τ , for each state $\mathbf{s}[t_\tau] \in \mathcal{S}$, the agent, i.e., the Hub, issues an action $\mathbf{a}[t_\tau] \in \mathcal{A}$ that maximizes the obtained reward $r[t_\tau]$.

Let $\gamma, 0 \leq \gamma \leq 1$, denote the *discount rate*. The *state-value function* for policy π , $V^\pi(\mathbf{s}) : \mathcal{S} \rightarrow \mathbb{R}$, which gives the expected return when starting from state $\mathbf{s}$ at time step t_τ and following policy π [10], is expressed as $V^\pi(\mathbf{s}) = \mathbb{E}_\pi[r(\mathbf{s}[t_\tau], \mathbf{a}[t_\tau]) + \gamma V^\pi(\mathbf{s}[t_{\tau+1}]) | \mathbf{s}[t_\tau] = \mathbf{s}]$. The *action-value function*, $q^\pi(\mathbf{s}, \mathbf{a}) : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, which is the expected return when starting from state $\mathbf{s} \in \mathcal{S}$, taking action $\mathbf{a} \in \mathcal{A}$ [10], is expressed as $q^\pi(\mathbf{s}, \mathbf{a}) = \mathbb{E}_\pi[v[t_\tau] | \mathbf{s}[t_\tau] = \mathbf{s}, \mathbf{a}[t_\tau] = \mathbf{a}]$. The aim of the agent is to find a policy π that gives the optimal action-value function [10], i.e., $q^*(\mathbf{s}, \mathbf{a}) = \max_\pi q^\pi(\mathbf{s}, \mathbf{a}), \forall \mathbf{s} \in \mathcal{S}, \forall \mathbf{a} \in \mathcal{A}$.

C. The Proposed DDQN Algorithm for the Equivalent MDP

In this work, we consider a practical scenario where the channel matrix $\mathbf{G}$ is unknown to the Hub. Thus, the transition probability in (15) is unavailable. Specifically, the considered environment of anti-reactive jamming is an *infinite-horizon non-episodic continuing-task* environment. Since the transition probability, which represents the dynamics of the environment in DRL, is unavailable to the agent, we resort to *temporal-difference (TD) learning*. Since the CFFA strategy changes

independently from one state to another, we rely on Q-learning, which is an *off-policy TD control*. Since the number of state and action pairs $(\mathbf{s}, \mathbf{a})$ is large in our network, it is computationally complex to calculate all Q-function values to determine the optimal policy. It is noted that DDQN is a model-free, online, off-policy DRL algorithm.

It is recalled that Λ denotes the number of time steps in one coherence time block. Thus, the faster the changes in channel conditions, the shorter T_c is, and the lower Λ is. The DDQN model is trained iteratively in each time step t_τ and converges to a stable reward value after a sufficient number of iterations. The input of the DDQN at time step t_τ is the observation of the MCFL, denoted by $\mathbf{o}[t_\tau]$, which consists of the state and real-valued channel magnitude as follows:

$$\mathbf{o}[t_\tau] = [s_{f_1}, \dots, s_{f_k}, \dots, s_{f_K}, h_{T_1, R_1}^{f_1}, \dots, h_{T_N, R_N}^{f_1}, \dots, h_{T_1, R_1}^{f_K}, \dots, h_{T_N, R_N}^{f_K}, f_J^*], \quad (16)$$

D. MDP-based Implementation of the Proposed Algorithm

1) *Action Selection based on Decaying ϵ -Greedy Policy*: In each iteration, an action $\mathbf{a} \in \mathcal{A}$ is selected based on the decay ϵ -greedy policy. Specifically, for exploring purposes, an action $\mathbf{a}$ is selected randomly from $\mathcal{A}$ with a probability ϵ . Otherwise, for exploiting purposes, an action $\mathbf{a}$ is selected based on the Bellman optimality equation as

$$\mathbf{a} = \arg \max_{\mathbf{a}_m \in \mathcal{A}} Q(\mathbf{s}', \mathbf{a}_m), \quad (17)$$

and the update of ϵ in each time step t_τ is based on the ϵ -greedy algorithm, which can be expressed as

$$\epsilon[t_\tau] = \max\{\epsilon_{\text{start}} - \kappa[t_\tau]\epsilon_{\text{dec}}, \epsilon_{\text{min}}\}, \quad (18)$$

where ϵ_{start} , ϵ_{dec} , ϵ_{min} , and κ are design hyper-parameters in Algorithm 1, and $\kappa[t_\tau] \triangleq (t_\tau \bmod T_c)$. We define $\epsilon_{\text{dec}} \triangleq 1/(\Lambda - \Lambda\beta\%)$, where β represents the portion of time steps used by the agent for exploitation purposes.

2) *Experience Replay*: After selecting an action $\mathbf{a}$, the corresponding immediate reward r and the next state $\mathbf{s}'$ are determined. Using the experience replay technique, an experience $(\mathbf{o}, \mathbf{a}, r, \mathbf{o}')$, where $\mathbf{o}$ is defined in (16), is stored in a replay buffer $\mathcal{B}$ as a data sample.

3) *Bellman Update at Each Iteration*: The DDQN algorithm is trained to iteratively adjust its weights θ to minimize the loss function, which is formulated as follows. The Q-function $Q(\mathbf{s}, \mathbf{a})$ is an estimation of the action-value function $q(\mathbf{s}, \mathbf{a})$ [10], which specifies how good is a particular action performed in a particular state with a particular policy. Thus, the optimal state-value function is expressed as $V^*(\mathbf{s}) = \max_\pi V^\pi(\mathbf{s}), \forall \mathbf{s} \in \mathcal{S}$. Consequently, the optimal action-value function is expressed as $Q^*(\mathbf{s}, \mathbf{a}) = \max_\pi Q^\pi(\mathbf{s}, \mathbf{a}), \forall \mathbf{s} \in \mathcal{S}, \forall \mathbf{a} \in \mathcal{A}$. It is noted that in our problem the agent performs off-policy learning, where it estimates the $q(\mathbf{s}, \mathbf{a})$ using the DQN, i.e., computes $Q(\mathbf{s}, \mathbf{a})$ to directly approximate the $q^*(\mathbf{s}, \mathbf{a})$. To estimate the value function, a DQN agent maintains two function approximators, namely $Q(\mathbf{s}[t_\tau], \mathbf{a}[t_\tau]; \theta[t_\tau])$ and $Q(\mathbf{s}[t], \mathbf{a}; \theta^-)$, which are

Algorithm 1: The DDQN-based CFFA algorithm.

Input : $M, K, \mathbf{H}, P_{T_n}, \sigma_{R_n}^2, \epsilon_{\text{start}}, \epsilon_{\text{dec}}, \epsilon_{\text{min}},$
 Batch, Epoch, $\text{lr}, M_{\text{max}}, \gamma, \nu_{\text{dqn}}, T_c$

Output: Action $\mathbf{a}$, immediate reward r

- 1 Generate network environment
 $\text{env} \leftarrow \text{NetworkEnv}(M, K, \mathbf{H}, P_{T_n}, \sigma_{R_n}^2)$
- 2 Initialize
 $\text{DQN}_{\text{core}}, \text{DQN}_{\text{RTR}}, \theta_{\text{DQN}_{\text{core}}}, \theta_{\text{DQN}_{\text{RTR}}} \leftarrow \text{DQNAgent}(\text{env})$
- 3 **for** each time block t with channel realization $\mathbf{H}(:, :, t)$
do
- 4 Randomly initialize $\mathbf{s}[0] \in \mathcal{S}, \mathbf{a}[0] \in \mathcal{A}$, set
 $\mathbf{s}'[0] \leftarrow \mathbf{0}, r[0] \leftarrow 0, i \leftarrow 1$
- 5 **for** $i \in \{1, 2, \dots, \nu_{\text{dqn}}\}$ **do** // Loop to train
 the DDQN model
- 6 Update $\mathbf{a}[i] \leftarrow \epsilon\text{-greedy}(\mathbf{s}[i-1])$
- 7 Obtain $\mathbf{s}'[i], r[i]$ from $\text{env.step}(\mathbf{a}[i])$;
 generate observations $\mathbf{o}[i]$ and $\mathbf{o}'[i]$; and store
 $(\mathbf{o}[i], \mathbf{a}[i], r[i], \mathbf{o}'[i])$ to the buffer $\mathcal{B}$
- 8 Randomly choose batches $\{\mathbf{o}_1, \dots, \mathbf{o}_B\},$
 $\{\mathbf{a}_1, \dots, \mathbf{a}_B\}, \{r_1, \dots, r_B\}, \{\mathbf{o}'_1, \dots, \mathbf{o}'_B\}$
- 9 Obtain
 $Q(\mathbf{s}', \mathbf{a}') \leftarrow \text{DQN}_{\text{core}}.\text{predict}(\{\mathbf{s}'_1, \dots, \mathbf{s}'_B\}),$
 and update $Q(\mathbf{s}, \mathbf{a})$ using (20)
- 10 Train DQN_{core} using
 $\text{DQN}_{\text{core}}.\text{fit}(\{\mathbf{s}_1, \dots, \mathbf{s}_B\}, Q(\mathbf{s}, \mathbf{a}))$
- 11 Decrease ϵ using (18), update $\mathbf{s}[i] \leftarrow \mathbf{s}'[i]$, save
 weights $\theta_{\text{DQN}_{\text{core}}}$
- 12 **end**
- 13 Update $\theta_{\text{DQN}_{\text{RTR}}}[\tau] \leftarrow \theta_{\text{DQN}_{\text{core}}}[\tau-1]$
- 14 Set $\tau \leftarrow 1$ and initialize
 $\mathbf{s}[\tau] = [\mathbf{t}_1, \dots, \mathbf{t}_N, \text{non}, \dots, \text{non}, f_1]$
- 15 **while** $\tau \leq T_c$ **do**
- 16 Predict $\mathbf{a}[\tau] \leftarrow \text{DQN}_{\text{RTR}}.\text{predict}(\mathbf{s}[\tau])$
- 17 Obtain $r[\tau], \mathbf{s}'[\tau] \leftarrow \text{env.step}(\mathbf{a}[\tau])$
- 18 Update $\mathbf{s}[\tau] \leftarrow \mathbf{s}'[\tau]$
- 19 **end**
- 20 **end**

refereed to as Q network and target Q network. At time step t_τ , the Q network $Q(\mathbf{s}[t_\tau], \mathbf{a}[t_\tau]; \theta[t_\tau])$ returns a *predicted* Q-value of the state-action pair $(\mathbf{s}[t_\tau], \mathbf{a}[t_\tau])$, while the target Q network returns a *target* Q-value, i.e.:

$$Q(\mathbf{s}[t_\tau], \mathbf{a}[t_\tau]; \theta^-) = r[t_\tau] + \gamma \max_{\mathbf{a}' \in \mathcal{A}} Q(\mathbf{s}[t_{\tau+1}], \mathbf{a}'; \theta). \quad (19)$$

Note that both training Q network and target Q network have the same structure and parameterization. The Bellman equation used to update the Q network from the target Q network is expressed as [10, Eq. (6.8)]

$$Q(\mathbf{s}[t_\tau], \mathbf{a}[t_\tau]) \leftarrow Q(\mathbf{s}[t_\tau], \mathbf{a}[t_\tau]) + \alpha \underbrace{(r[t_\tau] + \gamma \max_{\mathbf{a}' \in \mathcal{A}} Q(\mathbf{s}[t_{\tau+1}], \mathbf{a}'))}_{\text{target Q-value}} - \underbrace{Q(\mathbf{s}[t_\tau], \mathbf{a}[t_\tau])}_{\text{predicted Q-value}}, \quad (20)$$

where $0 < \alpha < 1$ is the *learning rate*.

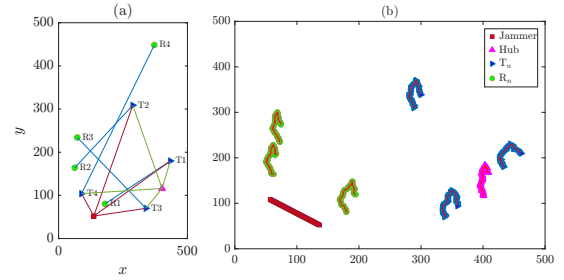
4) *Model Training:* For model training, we use the minimum squared error (MSE), i.e., the minimum squared difference of the target Q value and predicted Q value, as an objective function. Thus, the loss function, $\text{loss}(\theta[t_\tau])$, used in training $\theta[t_\tau]$, is defined as

$$\text{loss}(\theta[t_\tau]) = \mathbb{E}[(r[t_\tau] + \gamma \max_{\mathbf{a}' \in \mathcal{A}} Q(\mathbf{s}[t_{\tau+1}], \mathbf{a}'; \theta^-) - Q(\mathbf{s}[t_\tau], \mathbf{a}[t_\tau]; \theta[t_\tau]))^2]. \quad (21)$$

Next, the gradient decent method is performed to update the loss difference, i.e., $\nabla_{\theta} \text{loss}$.

IV. RESULTS AND DISCUSSIONS

In this section, we present representative results to discuss the performance of the proposed algorithm.



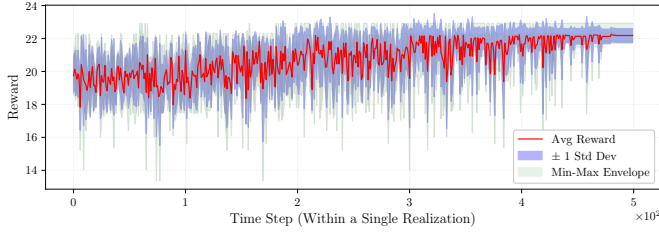


Fig. 2: Convergence of the DDQN model of the proposed algorithm considering one channel realization with $\beta = 5\%$, $N = 4$, $K = 5$, and $P_T = 0$ dBm.

2) *Numerical Results:* In Fig. 1(a), we present a snapshot of the MCFI network considered in our simulations. Specifically, all the user pairs, the Hub, and the jammer are heterogeneously deployed over the given square network area of 500×500 m², for the next time block, the Hub and users move following the RPGM model, while the jammer moves following the RWPM model, as depicted in Fig. 1(b). It is noted that each channel realization set, i.e., a set of $\mathbf{H}$ and $\mathbf{G}$, is simulated with respect to one snapshot of the network topology.

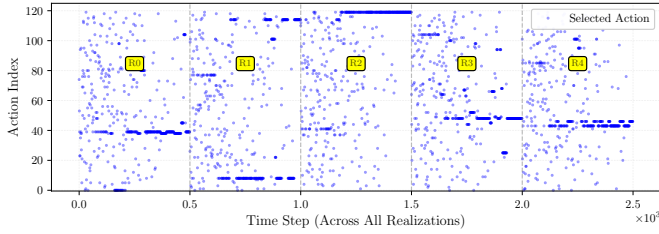


Fig. 3: Actions predicted by the DDQN algorithm considering one channel realization, with $\beta = 5\%$, $N = 4$, $K = 5$, and $P_T = 0$ dBm. R_i corresponds to time block i .

In Fig. 2, we present the convergence during the training phase of the DDQN algorithm. As observed in Fig. 2, after a series of trials, the algorithm converges to an optimal point, which returns the near-optimal value of the immediate reward. The converged optimal value obtained by the DDQN is approximately equal to the value found by the brute-force search algorithm, which validates the correctness of the proposed DDQN algorithms.

In Fig. 3, we demonstrate the exploration and exploitation of the proposed algorithm by plotting the action index of the action that the DDQN algorithm chooses in each time step. The speed of the convergence is determined by the decay rate of the ϵ value, ϵ_{dec} . As observed in Fig. 3, during the exploration phase, the algorithm selects various actions to explore more states in the state space; during the exploitation phase, only the action that returns the highest Q-value of that state-action pair is chosen for that state.

In Fig. 4, we compare the achievable SE obtained by the DDQN with that from the exhaustive search. As observed, user mobility has a significant impact on the achievable SE, and the DDQN reinitializes its learning for each new realization of

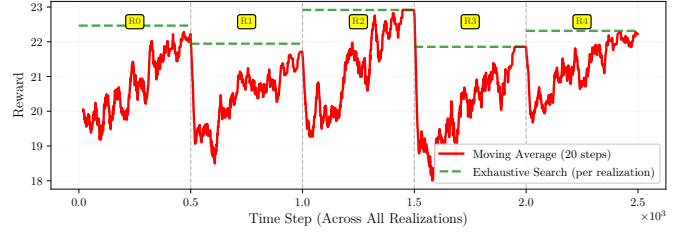


Fig. 4: Comparison against exhaustive search across different locations, with $\beta = 5\%$, $N = 4$, $K = 5$, and $P_T = 0$ dBm.

the MCFI network because it lacks prior knowledge (i.e., the CSI of the jammer) and instead relies on posterior knowledge, namely the jammed channel after the attack.

V. CONCLUSIONS

In this paper, we proposed an anti-reactive jamming CFFA strategy for MCFI networks aimed at enabling semi-distributed tactical transmissions over a jamming-free TCH. The design of the proposed CFFA strategy was formulated as an optimization problem under practical system constraints and subsequently modeled as a MDP. To solve this problem efficiently, we developed a DDQN-based learning algorithm. Numerical evaluations demonstrated the convergence behavior of the proposed method and confirmed its effectiveness by comparing its performance with that of an exhaustive search approach.

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